Distribution function Technique

Let $x_1, x_2 \cdots x_n$ be continuous random variables with a given joint probability density.

Let $Y = u(x_1, x_2 \cdots x_n)$, then the pdf of Y is obtained by first determining the cumulative distribution function (cdf) of Y, denoted by $F(y) = P(Y \leq y)$

$$= P[u(x_1 \cdots x_n) \leq y]$$

and then differentiate to get the pdf of Y ($f(y)$). i.e.

$$f(y) = \frac{d}{dy}[F(y)]$$

Example 1:

Let X be a random variable with pdf

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1 \\ 0, & \text{o.w} \end{cases}$$

Suppose $Y = X^3$, find the pdf of the random variable Y.

Soln

Step 1: Find the cdf of Y ($F(y)$).

$$F(y) = P(Y \leq y)$$

$$= P(X^3 \le y)$$
$$= P(X \le y^{1/3})$$
$$= \int_0^{y^{1/3}} 6x(1-x)\,dx$$
$$= 6\int_0^{y^{1/3}} (x - x^2)\,dx$$
$$= 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{y^{1/3}}$$
$$= 3y^{2/3} - 2y$$

$$\therefore \boxed{F(y) = 3y^{2/3} - 2y}$$

Step 2

To determine the pdf, use

$$f(y) = \frac{d}{dy} F(y)$$
$$= \frac{d}{dy}\left[3y^{2/3} - 2y\right]$$
$$= 2y^{-1/3} - 2$$
$$= 2(y^{-1/3} - 1)$$

$$\therefore \boxed{f(y) = \begin{cases} 2(y^{-1/3} - 1), & 0 < y < 1 \\ 0, & \text{o.w} \end{cases}}$$

Example

Let x has a pdf

$$f(x) = 3(1-x)^2, \quad 0 < x < 1.$$

Find the pdf of $Y = (1-X)^3$.

Solution

Using distribution function method.

It is clear that y decreases as x increases

Thus Y is a decreasing function of x and

$0 < y < 1$.

$$F(y) = Pr(Y \leq y)$$

$$= \int_{R_y} f_Y(x)\, dy$$

$$= \int_0^1 f_Y(x)\, dy.$$

but $x = 1 - y^{1/3}$.

$$\therefore F(y) = \int_0^y 3(1-(1-y^{1/3}))^2\, dy$$

$$= 3\int_0^y (y^{1/3})^2\, dy.$$

$$= y.$$

Hence

$$f(y) = y, \quad 0 \leq y < 1.$$

To find the pdf, we differentiate $F(y)$

$$\therefore f(y) = \frac{\partial}{\partial y} F(y) = \frac{\partial}{\partial y}[y] = 1.$$

Hence $f(y) = \begin{cases} 1, & 0 < y < 1 \\ 0, & 0.w. \end{cases}$

(5) Using Change of Variable approach.

$$f(y) = f_x(y)\,|J|.$$

Since $y = (1-x)^3 \Rightarrow x = 1 - y^{1/3}$.

Then

$$J = \frac{dx}{dy} = -\frac{1}{3} y^{-2/3}.$$

Therefore

$$f(y) = 3\left(1-(1-y^{1/3})\right)^2 \cdot \left|-\frac{1}{3} y^{-\frac{2}{3}}\right|$$

$$= 3y^{2/3} \cdot \frac{1}{3} y^{-2/3}$$

$$= 1$$

Hence

$$f(y) = \begin{cases} 1, & 0 < y < 1 \\ 0, & 0.w. \end{cases}$$

Example

Let X be a continuous random Variable with pdf

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{O.W.} \end{cases}$$

Find the probability of $Y = X^2$ using
(a) the distribution function approach
(b) the change of Variable approach

Solution

the graph of $y = x^2$ is an increasing function of X. and also $0 < y < 1$.

Using the distribution function, step 1 is to find the cdf.

$$F(y) = P_r(Y \le y)$$
$$= \int_{R_y} f(x)\,dy.$$

but $x = \sqrt{y} = y^{1/2}$.

$$\therefore F(y) = \int_0^y 3(y^{1/2})^2\,dy.$$
$$= 3\int_0^y (y^{1/2})^2\,dy$$
$$= 3\,\frac{y^{3/2}}{3} = y^{3/2}.$$

$$\therefore F(y) = y^{3/2}, \quad 0 < y < 1$$

Step 2:

$$f(y) = F'(y) = \frac{d}{dy} F(y).$$

$$= \frac{d}{dy}\left[y^{3/2} \right]$$

$$= \frac{3}{2} y^{1/2}$$

Hence the pdf of $Y = X^2$ is

$$f(y) = \begin{cases} \frac{3}{2} y^{1/2}, & 0 < y < 1 \\ 0. & \text{O.W.} \end{cases}$$

(b) Use the change of variable approach.

$$f(y) = f_X(y) \cdot |J|.$$

$$Y = X^2 \Rightarrow x = \sqrt{y} = y^{1/2}.$$

$$\frac{dx}{dy} = \frac{1}{2} y^{-1/2} = |J|.$$

$$\therefore f(y) = 3(y^{1/2})^2 \cdot \frac{1}{2} y^{-1/2}.$$

$$= \frac{3}{2} y^{1/2}.$$

Thus the pdf of $Y = X^2$ is

$$f(y) = \begin{cases} \frac{3}{2} y^{1/2}, & 0 < y < 1 \\ 0. & \text{O.W.} \end{cases}$$

Transformation Technique:

One Variable Case:

Let X and Y be two Independent Variables with a one-to-one relationship that $Y = u(X)$.

For a discrete case, the determination of the pdf of Y from X is straight forward.

Example

Let X be the number of heads obtained in four tosses of a balanced coin, find the probability distribution of $Y = \frac{1}{1+X}$.

Solution

$X \sim B(n, \theta)$, where $\theta = 1/2$, $x = 0, 1, 2, 3, 4$.

$P(X = x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$.

$\therefore P(x=0) = \binom{n}{0} \theta^0 (1-\theta)^{n-0} = (1/2)^4 = 1/16$,

$P(x=1) = \binom{4}{1} \theta (1-\theta)^3 = 4/16$

$P(x=2) = \binom{4}{2} \theta^2 (1-\theta)^2 = 6/16$

$P(x=3) = \binom{4}{3} \theta^3 (1-\theta) = 4/16$.

$P(x=4) = \binom{4}{4} \theta^4 (1-\theta)^0 = 1/16$.

Thus

x	0	1	2	3	4
$f(x)$	1/16	4/16	6/16	4/16	1/16

Then, to find the Pdf of Y, we use the relationship $y = \frac{1}{1+x}$.

Thus the range of $y = 1, 1/2, 1/3, 1/4, 1/5$

$\therefore g(y) = \left\{\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}\right\}$.

Note that $y = \frac{1}{1+x} \Rightarrow x = \frac{1}{y} - 1$.

Thus

$$g(y) = f\left(\frac{1}{y} - 1\right) = \binom{4}{\frac{1}{y}-1}\left(\frac{1}{2}\right)^{1/y-1}\left(\frac{1}{2}\right)^{n-\left(\frac{1}{y}-1\right)}.$$

$$= \binom{4}{\frac{1}{y}-1}\left(\frac{1}{2}\right)^{n}; \quad y = 1, 1/2, 1/3 \ldots$$

Note that if the relationship is not one-one, the approach of transformation (otherwise called change of Variable) will be different.

Example (non one-one Relationship).

Let X be the number of heads obtained in four tosses of a balance (fair) coin. Suppose that $Y = (X-2)^2$, find the Pdf of Y.

Soln.

$X \sim B(n, \theta) \Rightarrow n = 4, \theta = 1/2, x = 0, 1, 2, 3, 4.$

$h(y) = h\left((x-2)^2\right) \Rightarrow \boxed{x = y^{1/2} + 2}$

$\therefore h(0)$

Thus when $x = 0 \Rightarrow y = 4$.
$x = 1 \Rightarrow y = 1$
$x = 2, \quad y = 0$
$x = 3, \quad y = 1$
$x = 4, \quad y = 4$

$\therefore R_y = 0, 1, 4.$

The probability distribution function of y are

$h(0) = f(2) = 6/16.$

$h(1) = f(1) + f(3) = 4/16 + \frac{4}{16} = \frac{8}{16} = 1/2$

$h(4) = f(0) + f(4) = 1/16 + 1/16 = \frac{2}{16} = \frac{1}{8}.$

For Continous Case

The transformation technique for continous case is based on the assumption that the function given by ~~$y = f(x)$~~

$y = u(x)$ is differentiable and its monotonically increasing or decreasing for all values within the range of x for which $f(x) \neq 0$.

Second, it is assumed that $x = w(y)$ exists for all corresponding values of y and it is differentiable.

Example

If x has the exponential distribution with pdf

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{o.w.} \end{cases}$$

Find the probability distribution of the random variable $Y = \sqrt{X}$.

Solution

$$Y = \sqrt{X} \Rightarrow x = y^2.$$

Thus $\frac{dx}{dy} = 2y$. Therefore

$$g(y) = e^{-y^2} |J|, \text{ where } J = \frac{dx}{dy} \text{ which is called the Jacobian of the transformation}$$

$$\therefore g(y) = e^{-y^2} |2y|$$

$$= \begin{cases} 2ye^{-y^2}, & y > 0. \\ 0, & \text{o.w.} \end{cases}$$

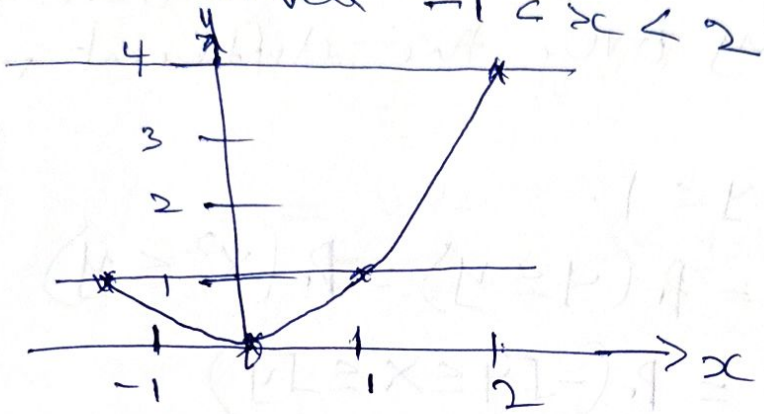
Exercise

Let x be a random variable from the Standard normal distribution, Find the probability density function of $Z = x^2$.

Two-to-One-Functions

When the relationship between two variables are non-monotonic, the transformation approach will have to be modified.

For example $y = x^2$ is not one-to-one over the interval $-1 < x < 2$.



Note that the inverse here will be $x = y^{1/2}$.

For instance, in the interval $-1 < x < 1$, if we take the inverse of $y = x^2$, we get

$$x_1 = -\sqrt{y} \quad \text{for } -1 < x < 0$$

Also if we take the interval $0 < x < 2$, we get

$$x_2 = +\sqrt{y}.$$

Thus there is two segment for the range of y. First $0 \leq y < 1$ and also $1 \leq y < 4$

Example

Let x be continuous random variable with pdf

$$f(x) = \frac{x^2}{3}, \quad -1 < x < 2$$

Find the pdf of $y = x^2$.

Soln

It has been shown that the graph of the function y is a non-linear function. Also the range of y is divided into two segment as shown the graph.

(a) Using distribution function approach, we have to find the distribution function separately over the different ranges of y.

for $0 < y < 1$.

$$F(y) = P_r(Y \le y) = P_r(X^2 \le y)$$
$$= P_r(-\sqrt{y} \le X \le \sqrt{y})$$
$$= F_x(\sqrt{y}) - F_x(-\sqrt{y})$$

but $f(x) = x^2/3$.

Then $\Rightarrow y = x^2 \Rightarrow x = \sqrt{y} = y^{1/2}$

$$F[y] = F(y^{1/2}) - F(-y^{1/2}).$$

The pdf of y is

$$f(y) = \frac{d}{dy} F(y)$$
$$= \frac{(\sqrt{y})^2}{3} \cdot \frac{1}{2} y^{-1/2} + \frac{(-\sqrt{y})^2}{3} \cdot \frac{1}{2} y^{-1/2}$$
$$= \frac{1}{6} y^{1/2} + \frac{1}{6} y^{1/2} = \frac{\sqrt{y}}{3} = \frac{y^{1/2}}{3}.$$

Hence $f(y) = \begin{cases} \frac{y^{1/3}}{3}, & 0 < y < 1 \\ 0, & \text{o.w} \end{cases}$

for $1 < y < 4$ we have

$$F(y) = P(Y \le y) = P(X^2 \le y) = P(X \le \sqrt{y})$$
$$= F_X(\sqrt{y}).$$

Thus

$$f(y) = \frac{d}{dy} F(y) = f_X(\sqrt{y}) \cdot \tfrac{1}{2} y^{-1/2}.$$
$$= \frac{(\sqrt{y})^2}{3} \cdot \tfrac{1}{2} y^{-1/2}$$
$$= \frac{y^{1/2}}{6}, \quad 1 < y < 4.$$

Theorem

Let X be a continuous r.v with pdf $f(x)$, $c_1 < x < c_2$. Let $Y = u(X)$ be a continuous two-one function of X, which can be splitted into two one-one invertible functions with

$$x_1 = v_1(y) \text{ and } x_2 = v_2(y).$$

Then

(i) the pdf for the two-one portion of Y is

$$f_Y(y) = f_X(v_1(y)) \cdot |v_1'(y)| + f_X(v_2(y)) \cdot |v_2'(y)|$$

(ii) the pdf for the one-one portion of Y is

$$f_Y(y) = f_X(v_2(y)) \cdot |v_2'(y)|.$$

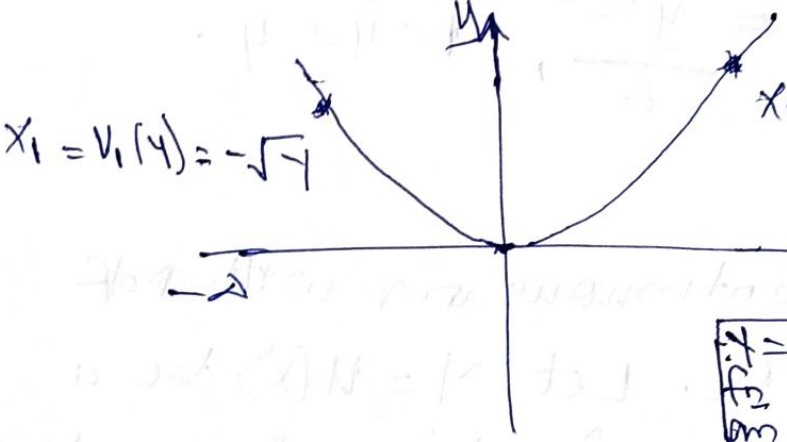
Example

$X \sim N(0,1)$. $-\infty < x < \infty$

Show that the pdf of $Y = X^2$ is the Chi-Square distribution with 1 degree of freedom.

Soln

$Y = X^2$ is a two-one function over the range of X.



$Y = x^2$ is a decreasing fn for -ve value of x and an increasing fn for +ve value.

Thus when $-\infty < x < 0$, we have

$X_1 = -\sqrt{Y} = V_1(y)$ and when

$0 < x < \infty$, we have

$X_2 = \sqrt{Y} = V_2(y)$

Then, over the two-one portion of the transformation, i.e. when $0 < y < \infty$

$$f_Y(y) = f_X(\sqrt{y}) \left| \frac{1}{2} y^{-1/2} \right| + f_X(-\sqrt{y}) \cdot \left| -\frac{1}{2} y^{-1/2} \right|$$

But $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$,

Then

$$f(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\sqrt{y})^2} \cdot \frac{1}{2} y^{-1/2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-\sqrt{y})^2} \cdot \frac{1}{2} y^{-1/2}$$

$$= \frac{2}{\sqrt{2\pi}} e^{-\frac{y}{2}} \cdot \frac{1}{2} y^{-1/2}.$$

but $\sqrt{\pi} = \Gamma(1/2)$ and $\sqrt{2} = 2^{1/2}$.

$$\therefore f(y) = \frac{1}{\Gamma(1/2)\, 2^{1/2}} e^{-y/2} y^{-1/2} \sim \chi^2_1.$$

Thus the pdf of y is

$$f(y) = \begin{cases} \frac{1}{\Gamma(1/2)\, 2^{1/2}} e^{-y/2} y^{-1/2}, & 0 < y < \infty \\ 0, & \text{o.w.} \end{cases}$$

Transformation Technique: Two-Dimensional Random Variables

Suppose (X, Y) is a two-dimensional Continuous Random Variables with joint Pdf $f(x, y)$.

Let $U = G_1(X, Y)$ and $V = G_2(X, Y)$, where G_1 and G_2 satisfy the followup:

(i) $u = G_1(x, y)$, $v = G_2(x, y)$ may be Uniquely solved for x and y in terms of U and V only i.e. $x = W_1(U, V)$, $y = W_2(U, V)$

(ii) $\frac{\partial x}{\partial u}$, $\frac{\partial x}{\partial v}$, $\frac{\partial y}{\partial u}$, and $\frac{\partial y}{\partial v}$ exist and are Continuous.

Then the joint pdf of U and V is given by

$$g(U, V) = f(W_1(U, V), W_2(U, V)) |J(U, V)|$$

where $J(U, V)$ is the Jacobian of transformation and

$$J(U, V) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

It should be noted that the marginal pdf for U and V can be obtained from $g(U, V)$.

Example

Suppose X and Y are independent random variables of the continuous type with joint pdf

$$f(x,y) = e^{-(x+y)}, \quad x>0;\ y>0.$$

Let $U = X+Y$, find the pdf of U.

Solution

$$U = X+Y \rightarrow G_1(x,y)$$

Let $V = X \longrightarrow G_2(x,y)$.

Then

$$x = V \rightarrow W_1(u,v), \quad \frac{dx}{dv} = 1, \ \frac{dx}{du} = 0$$

and

$$y = u-v \rightarrow W_2(u,v). \quad \frac{dy}{dv} = -1, \ \frac{dy}{du} = 1.$$

Therefore; The Jacobian of the transformation

$$J(u,v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1.$$

$$\therefore |J(u,v)| = 1.$$

Then the joint pdf of U and V is

$$g(u,v) = f(W_1(u,v), W_2(u,v))\,|J(u,v)|.$$

$$= e^{-(v+u-v)} * 1$$

$$= e^{-u}, \quad u>0,\ v<u.$$

To find the marginal pdf of u, we integrate the joint pdf.

Thus

$$f(u) = \int_{R_u} g(u,v)\,dv$$

$$= \int_0^u e^{-u}\,dv.$$

$$= e^{-u}\int_0^u dv = e^{-u}[u-0]$$

$$= ue^{-u}, \quad u > 0.$$

Hence

$$f(u) = \begin{cases} ue^{-u}, & u > 0 \\ 0, & 0.w \end{cases}$$

Example (Differences)

$$f(x,y) = e^{-(x-y)} \quad \begin{matrix} x>0 \\ y>0 \end{matrix}$$

Suppose $u = x - y$.

Find the pdf of u.

Solution.

$$u = x - y \longrightarrow G_1(x,y)$$

$$v = x \longrightarrow G_2(x,y)$$

It implies that

$$x = v \quad \ldots\ldots (i)$$

$$y = v - u \quad \ldots\ldots (ii)$$

Then

$$J(u,v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = 1$$

$\therefore |J(u,v)| = 1$.

Thus

$$g(u,v) = e^{-(u+(v-u))} * 1$$
$$= e^{-u}, \quad u > 0, \; v < u.$$

$$\therefore g(u,v) = \begin{cases} e^{-u}, & u > 0, \; v < u. \\ 0, & \text{o.w.} \end{cases}$$

To find the pdf of u, we solve

$$f(u) = \int_0^{\infty} g(u,v)\, dv$$
$$= \int_0^{\infty} e^{-u}\, dv$$
$$= e^{-u}\left[v\right]_0^{\infty}$$
$$= -ue^{-u}, \quad u > 0.$$

Hence

$$f(u) = \begin{cases} -ue^{-u}, & u > 0 \\ 0, & \text{o.w.} \end{cases}$$

Example

Let U be a standard normal random variable and V a chi-square random variable with m degree of freedom. Assuming that U and V are independent, find the pdf of

$$T = \frac{U}{\sqrt{V/m}}.$$

Solution

$$t = \frac{u}{\sqrt{v/m}} \Rightarrow u = t\sqrt{v/m}$$

$U \sim N(0,1)$

$V \sim \chi^2_m$.

Define $W = V$ so that

$$v = w \quad \text{and} \quad u = t\sqrt{\frac{w}{m}}$$

Then the Jacobian of the transformation will be

$$J(u,v) = \begin{vmatrix} \frac{\partial u}{\partial t} & \frac{\partial u}{\partial w} \\ \frac{\partial v}{\partial t} & \frac{\partial v}{\partial w} \end{vmatrix} = \begin{vmatrix} \sqrt{\frac{w}{m}} & \frac{t}{2\sqrt{mw}} \\ 0 & 1 \end{vmatrix} = \sqrt{\frac{w}{m}}$$

$$\therefore f(u,v) = f(u)\cdot f(v)$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \cdot \frac{1}{\Gamma(m/2)\,2^{m/2}} v^{\frac{m}{2}-1} e^{-\frac{v}{2}} \left(\frac{v}{m}\right)^{1/2}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2 v}{2m}} \cdot \frac{1}{\Gamma(m/2)\,2^{m/2}} v^{\frac{m}{2}-1} e^{-\frac{v}{2}}$$

$$= \frac{1}{\sqrt{2\pi}\,\Gamma(m/2)\,2^{m/2}} v^{\frac{m}{2}-1} e^{-\frac{1}{2}\left(\frac{t^2}{v}+v\right)}.$$

$$= \frac{1}{\sqrt{2\pi}\,\Gamma(m/2)\,2^{\frac{m}{2}}}\, v^{\frac{m}{2}-1} e^{-\frac{1}{2}\left(1+\frac{t^2}{m}\right)v}.$$

Thus

$$g(t,w) = \frac{1}{\sqrt{2\pi}\,\Gamma(m/2)\,2^{m/2}}\, v^{\frac{m}{2}-1} e^{-\frac{1}{2}\left(1+\frac{t^2}{m}\right)v} \cdot \sqrt{\frac{w}{m}}$$

The marginal pdf of T becomes

$$h(t) = \int_0^\infty g(t,w)\,dw$$

$$= \frac{1}{\sqrt{2\pi}\,\sqrt{m}\,\Gamma(m/2)\,2^{\frac{m+1}{2}}} \int_0^\infty w^{\frac{m+1}{2}-1} e^{-\frac{w}{2}\left(1+\frac{t^2}{m}\right)} dw$$

$$= \frac{1}{\sqrt{\pi}\sqrt{m}\,\Gamma(m/2)\,2^{\frac{m+1}{2}}} \int w^{\frac{m+1}{2}-1} e^{-\left(\frac{\frac{w}{2}}{1+\frac{t^2}{m}}\right)} dw$$

$$= \frac{1}{\sqrt{m}\,\Gamma(m/2)\,2^{\frac{m+1}{2}}}\, \Gamma\left(\frac{m+1}{2}\right)\left(\frac{2}{1+\frac{t^2}{m}}\right)^{\frac{m+1}{2}}.$$

$$= \frac{1}{\sqrt{m}\,B\left(\frac{1}{2},\frac{m}{2}\right)\left(1+\frac{t^2}{2}\right)^{\frac{m+1}{2}}}, \quad -\infty < t < \infty.$$

It should be noted that $\sqrt{\pi} = \Gamma(1/2)$

Assignment 1